E-Commerce Analytics Platform

Revenue Optimization & Customer Intelligence

End-to-end analytics pipeline — MySQL · Python · Power BI

Project Overview

A fast-growing e-commerce company faced rising customer acquisition costs, a 68% cart abandonment rate, 22% product return rate, frequent inventory stockouts, and only 78% on-time delivery rate. This project builds a centralized E-Commerce Analytics Platform providing 360° visibility across customer behaviour, sales performance, marketing effectiveness, inventory movement, and logistics operations.

50,000	5,000	1,000	8	3
Sales Records	Customers	Products	Phases	ML Models

Table of Contents

Phase 1

Database Design & ETL

Star schema architecture · SCD Type 2 · 7 tables · MySQL Workbench

Architecture

Designed a star schema data warehouse with one central fact table and six dimension tables. Applied Slowly Changing Dimension Type 2 on dim_customer to track historical changes in customer attributes. The dim_date table enables time intelligence for MoM, YoY, and quarterly analysis.

Dataset Summary

Table	Source File	Rows	Key Columns
fact_sales	Sales.csv	50,000	Order_ID, Customer_Key, Product_Key, Net_Amount, Return_Flag
dim_customer	Customer.csv	5,000	Customer_Key, Region, Loyalty_Tier, CLV, Age_Band
dim_product	Product.csv	1,000	Product_Key, Category, Brand, Unit_Price, Cost_Price
dim_delivery	Delivery.csv	50,000	Delivery_ID, Partner_Name, Actual_Delivery_Days, Delivery_Status
dim_inventory	Inventory.csv	13,250	Product_Key, Warehouse_ID, Closing_Stock, Stockout_Flag
dim_marketing	Marketing.csv	120	Campaign_ID, Channel, Spend_Amount, Revenue_Generated, CAC
dim_date	Date.csv	365	Date_Key, Full_Date, Year, Quarter, Month_Name

Key SQL — Table Creation

```
CREATE TABLE fact_sales (
Sales_Key INT PRIMARY KEY,
Order_ID VARCHAR(20) NOT NULL,
Customer_Key INT,
Product_Key INT,
Net_Amount DECIMAL(14,2),
Return_Flag TINYINT(1),
CONSTRAINT fk_customer FOREIGN KEY (Customer_Key)
REFERENCES dim_customer(Customer_Key)
);
```

ETL Process

- Handled missing values in customer location and delivery time fields
- Removed duplicate orders and customer records
- Validated pricing, discount, and shipping cost fields
- Converted date strings to DATE format across all 7 tables
- Loaded dimensions first, fact table last to respect foreign key constraints

- SCD Type 2 columns added: scd_start_date, scd_end_date, scd_is_current

Phase 2

SQL Analytics

CTEs · Window Functions · Subqueries · Business Intelligence Queries

Key Queries Performed

CLV & Ranking	Top 10 customers by Customer Lifetime Value ranked within each region using RANKX and window functions
Product Risk	Products scored on weighted delivery delay, discount level, and order volume — normalised 0 to 1
MoM Revenue	Month-over-month revenue growth calculated using LAG() window function to flag revenue decline months
Stockout Impact	Products below reorder level identified and correlated with sales performance loss
Marketing ROI	ROI, conversion rate, and CAC computed per channel to identify best and worst performing campaigns

Sample Query — Month-over-Month Revenue Growth

```
WITH monthly_revenue AS (
SELECT DATE_FORMAT(Order_Date, '%Y-%m') AS month,
SUM(Net_Amount) AS revenue
FROM fact_sales
GROUP BY DATE_FORMAT(Order_Date, '%Y-%m')
),
mom_growth AS (
SELECT month, revenue,
LAG(revenue) OVER (ORDER BY month) AS prev_revenue,
ROUND((revenue - LAG(revenue) OVER
(ORDER BY month)) / LAG(revenue)
OVER (ORDER BY month) * 100, 2) AS growth_pct
FROM monthly_revenue
)
SELECT * FROM mom_growth
WHERE growth_pct < 0
ORDER BY month;
```

Phase 3

Statistical Analysis

Hypothesis Testing · ANOVA · Correlation · Python / Jupyter

Statistical Tests Performed

Test	Variables	Result	P-Value	Conclusion
T-Test (Hypothesis)	Delivery Delay vs Return Rate	Delayed Return Rate higher	< 0.05	Delays increase returns
ANOVA	Revenue across 6 Marketing Channels	F-stat: 1.4686	0.2056	No significant difference
Pearson Correlation	Delivery Days vs Return Flag	r = -0.0035	—	No linear correlation
Courier Analysis	Partner Name vs Return Rate	All partners 22%	—	Systemic, not partner issue

Key Findings

- Delivery delays DO increase return probability — confirmed by T-test ($p < 0.05$)
- No single courier partner is underperforming — all show identical 22% return rate
- Revenue does not differ significantly across marketing channels ($p = 0.2056$)
- Pearson $r = -0.0035$ confirms return driver is categorical (Delivery_Status), not linear
- All couriers average 7.4 delivery days against SLA targets of 4-6 days — systemic pre-dispatch issue

Phase 4

Power BI Dashboards

DAX Measures · Drill-through · What-If Parameters · 6 Pages

Dashboard Pages

Page	Title	Key Visuals	DAX Measures
1	Customer Intelligence	Bar chart top 10 CLV, table, region slicer	CLV, Customer_Rank
2	Product Risk Analysis	Risk score bar chart, conditional formatting, table	Product_Risk_Score, Risk_Label
3	Revenue Trends	Line chart MoM, year/quarter slicer, cards	MoM_Revenue_Growth %
4	Inventory & Stockouts	Stockout bar chart, warehouse slicer, table	Stockout_Products, Stockout_Sales_Impact
5	Marketing Performance	ROI by channel, spend vs revenue, CAC table	Marketing_ROI %, CAC, Conversion_Rate %
6	ML Predictions & Alerts	30-day forecast, churn table, return risk table	High_Churn_Customers, High_Return_Products

Key DAX Measures

```
-- Product Risk Score (normalised 0-1)
Product_Risk_Score =
VAR avg_delay = CALCULATE(AVERAGE(dim_delivery[Actual_Delivery_Days]))
VAR avg_discount = CALCULATE(AVERAGE(fact_sales[Discount_Amount]))
VAR avg_qty = CALCULATE(AVERAGE(fact_sales[Quantity]))
VAR max_delay = CALCULATE(MAX(dim_delivery[Actual_Delivery_Days]),
ALL(dim_delivery))
VAR max_discount = CALCULATE(MAX(fact_sales[Discount_Amount]),
ALL(fact_sales))
VAR max_qty = CALCULATE(MAX(fact_sales[Quantity]), ALL(fact_sales))
RETURN ROUND(
(DIVIDE(avg_delay,max_delay,0) * 0.4) +
(DIVIDE(avg_discount,max_discount,0) * 0.3) +
(DIVIDE(avg_qty,max_qty,0) * 0.3), 2)
```

```
-- Marketing ROI %
Marketing_ROI % =
DIVIDE(
SUM(dim_marketing[Revenue_Generated])
- SUM(dim_marketing[Spend_Amount]),
SUM(dim_marketing[Spend_Amount]), 0) * 100
```

Phase 5

Predictive ML Modeling

Sales Forecasting · Churn Prediction · Return Prediction · Python

Model Summary

Use Case	Model	Algorithm	Key Metric	Result
Sales Forecasting	Time Series	Prophet	MAPE	13.51%
Churn Prediction	Classification	Logistic Regression	ROC-AUC	0.9997
Churn Prediction	Classification	Random Forest	ROC-AUC	1.0000
Return Prediction	Classification	Logistic Regression	ROC-AUC	1.0000
Return Prediction	Classification	Random Forest	ROC-AUC	1.0000

Model 1 — Sales Forecasting (Prophet)

- Trained on daily aggregated Net_Amount from fact_sales
- Last 30 days held out as test set — MAPE: 13.51%
- Clear upward trend identified — revenue growing consistently through 2025
- Weekly seasonality: Saturday highest, Sunday/Tuesday lowest
- Yearly seasonality: December spike, August dip

Model 2 — Customer Churn Prediction

- Churn defined as no purchase in last 90 days — 4.46% churn rate (223 of 5,000 customers)
- Features: Total_Orders, Total_Revenue, Avg_Order_Value, Return_Rate, demographics
- Recency_Days excluded from features to avoid data leakage
- Random Forest outperformed Logistic Regression — selected as final model
- 57 customers flagged as High churn risk (probability >= 0.70)

Model 3 — Return Prediction

- Return rate: 22% across 50,000 orders
- Delivery_Status excluded from features as it directly encodes the outcome
- Top features: Discount_Amount, Sales_Amount, Actual_Delivery_Days
- Both models achieve strong classification — Random Forest selected
- 6,000+ orders flagged as High return risk for proactive intervention

Phase 6

BI Integration & Early Warning

Predictions pushed to MySQL · Power BI live alerts · 3 prediction tables

Prediction Tables Loaded into MySQL

Table	Rows	Key Columns	Purpose
ml_sales_forecast	30	Forecast_Date, Forecasted_Sales, Lower_Bound, Upper_Bound	30-day revenue forecast
ml_churn_predictions	5,000	Customer_Key, Churn_Probability, Risk_Level	Flag high-risk customers
ml_return_predictions	50,000	Order_ID, Return_Probability, Return_Risk	Flag high-risk orders

Early Warning Dashboard

- 57 High Churn Risk customers visible in real-time with probability scores
- 6K High Return Risk orders flagged with average return probability per product
- Forecasted revenue for next 30 days: 100.40M with confidence interval
- All 3 prediction tables connected to Power BI via direct MySQL query
- Drill-through enabled from any dashboard page to ML predictions page

Phase 7

Strategic Recommendations

Data-driven actions across Customer · Marketing · Inventory · Logistics · Returns

1. Customer Strategy

- Launch reactivation campaign targeting 57 high-risk churn customers immediately
- Tier-based loyalty rewards to move Bronze customers up the CLV ladder
- 90-day re-engagement rule — automated nudge at 60 days before churn threshold
- Personalized product recommendations based on past purchase category and region

2. Marketing Optimization

- Shift budget to Email and Influencer — highest avg revenue (16.6M and 16.1M)
- Reduce Push Notification spend — lowest revenue at 8.8M, no statistical justification
- Set CAC ceiling per channel — pause campaigns exceeding 2x blended CAC of 8,220
- ANOVA confirms no significant revenue difference across channels — optimise by ROI not revenue

3. Inventory Strategy

- Demand-driven replenishment using Prophet forecast to predict weekly demand per category
- Dynamic safety stock: avg lead time x avg daily demand + buffer — replace static levels
- Monthly SKU rationalisation — drop low velocity, high stockout products
- Priority focus on Electronics and Fashion — highest return volume categories

4. Logistics Improvements

- Revise SLA targets from 4-6 days to 7 days reflecting actual 7.4 day avg delivery
- Audit pre-dispatch time — all 4 couriers perform equally, bottleneck is warehouse side
- Show realistic delivery window at checkout using ML prediction model
- Pilot express delivery for high CLV customers via one selected partner

5. Returns Reduction

- Proactive outreach for orders with return probability > 0.70 before delivery
- Improve product detail pages — enhanced images, sizing guides, specifications
- Stricter quality checks for Electronics and Fashion categories
- Add return reason codes to dataset — will dramatically improve future model accuracy

Target KPIs — 6 Month Outlook

Metric	Current	Target (6 months)
Monthly Revenue	~2.5M avg	3.2M (Prophet forecast)
Churn Rate	4.46%	Below 3%
Return Rate	22%	Below 15%
On-Time Delivery	~78%	Above 90%
Customer Acq. Cost	8,220	Below 6,000

Tech Stack

Database	MySQL 8.0 — Star schema, foreign keys, indexes, SCD Type 2
ETL	Python — pandas, SQLAlchemy, pymysql
Analytics	MySQL Workbench — CTEs, Window Functions, Subqueries
Visualization	Power BI Desktop — DAX, drill-through, what-if parameters
ML / Stats	Python — scikit-learn, Prophet, scipy, matplotlib, seaborn
Models	Prophet (forecasting), Logistic Regression, Random Forest